Equações do indicador de otimismo/pessimismo

Carlos Viana, Marlon Koga

2020-11-16

$$\begin{bmatrix}1 & PRE & IBOV & SPX & USD\end{bmatrix}\begin{bmatrix}x_0\\x_{PRE}\\x_{IBOV}\\x_{SPX}\\x_{USD}\end{bmatrix} = b + \epsilon \tag{1}$$

Onde:
b - Retorno do fundo $[n \times 1]$
ϵ - Erro da regressão $[n \times 1]$

$$\begin{bmatrix}1 & PRE & [A]\end{bmatrix}\begin{bmatrix}x_0\\x_{PRE}\\\{x\}\end{bmatrix} = b + \epsilon \tag{2}$$

Tal que:
A - Matriz que agrupa os retornos do IBOV, SPX e USD $[3 \times 1]$

$$\begin{bmatrix}1 & PRE & [A]\end{bmatrix}\begin{bmatrix}x_0\\x_{PRE}\\(x-\bar{x})\end{bmatrix} = b - A\bar{x} + \epsilon \tag{3}$$

$$\begin{bmatrix}1 & PRE & A.diag(\sigma)\end{bmatrix}\begin{bmatrix}x_0\\x_{PRE}\\(\frac{x-\bar{x}}{\sigma})\end{bmatrix} = b - A\bar{x} + \epsilon \equiv b^* + \epsilon \tag{4}$$

$$\begin{bmatrix}1 & PRE & A.diag(\sigma)w\end{bmatrix}\begin{bmatrix}x_0\\x_{PRE}\\\beta\end{bmatrix} = b - A\bar{x} + \epsilon \tag{5}$$

$$A^*x^* = b^* + \epsilon \tag{6}$$

$$\begin{bmatrix}1 & PRE & IBOV & SPX & USD\end{bmatrix}\begin{bmatrix}x_0\\x_{PRE}\\diag(\sigma)w\beta + \bar{x}\end{bmatrix} = b + \epsilon \tag{7}$$

$$w = [-\alpha_{pre}^2] \tag{8}$$